

21 D D_1 is reversed biased, so pd across it is $-E$ and pd across R_1 is 0 V

22 D For steady current,

$$P = I^2 R$$

For half – wave rectified ac,

$$P = \left(\frac{I_o}{2}\right)^2 \left(\frac{1}{2}R\right) = I^2 R$$

$$I_o = \sqrt{8I} = 2.8I$$

23 B By right hand grip rule, magnetic flux density at Q due to B and C points in direction of C